

From Fair Shares to Carbon Debt: Quantifying National Responsibility and Policy Gaps under the Paris Agreement

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Interactive Dashboard: [Carbon Budget Tracker Deployment](#)

Code Repository: [GitHub: nishaverma24-bot/carbon-budget-tracker](#)

Abstract

Traditional climate accounting metrics frequently overemphasize current annual or absolute national emissions pathways, masking historical responsibilities and ethical resource misallocations. This paper presents the *Carbon Budget Tracker*, an open-science computational framework that downscales the Intergovernmental Panel on Climate Change (IPCC) Sixth Assessment Report (AR6) remaining carbon budgets using a dynamic, equal per-capita equity filter. By programmatically harmonizing multi-source historical datasets from Our World in Data (OWID), the World Bank, and the UNFCCC, the tracker computes national "Carbon Debt," "Justice Scores," and empirical "Policy Gaps." The framework demonstrates that 51 nations have completely exhausted their fair-share allocations, led by the United States (136.1 Gt carbon debt), while key developing economies like India remain major carbon creditors (-151.4 Gt).

Furthermore, this paper introduces diagnostic stress-testing modules, decomposing the structural drivers of overshoot (e.g., an industrialization age of 231 years for legacy economies) and assessing emergent digital vulnerabilities, such as AI-driven electricity demand spikes. By incorporating explicit methodological caveats regarding baseline selection and allocation protocols, this platform provides policymakers with a rigorous, transparent tool to bridge data science, climate equity, and systemic policy reform.

1. Introduction

The Paris Agreement is built on the principle of nationally determined climate action. Countries define their own mitigation pathways through Nationally Determined Contributions (NDCs), while global progress is commonly assessed through emissions trajectories, temperature projections, and net-zero commitments¹. Although these metrics provide valuable insights into future climate outcomes, they often overlook a fundamental constraint underlying the entire climate system: the remaining global carbon budget. The concept of a carbon budget recognizes that cumulative carbon dioxide emissions are the primary determinant of long-term global warming². Because the atmosphere represents a finite shared resource, questions of allocation inevitably emerge. How much of the remaining carbon budget should each country be entitled to use? Which nations have already exceeded their equitable share? And how should historical responsibility be incorporated into future climate governance discussions? Most existing climate dashboards focus on emissions flows rather than atmospheric stock consumption. While they effectively track annual emissions, they rarely quantify how much of the remaining carbon budget has already been consumed by individual countries relative to an equity-based allocation framework. As a result, important questions regarding climate justice, historical responsibility, and carbon debt remain difficult to assess in a transparent and reproducible manner. To address this gap, this paper introduces the Carbon Budget Tracker, an open-science computational framework designed to translate global carbon budgets into country-level indicators of responsibility, overshoot, and policy alignment. Using remaining carbon budgets reported by the IPCC Sixth Assessment Report (AR6)³, the framework allocates atmospheric space according to an Equal Per-Capita (EPC)⁴ principle and continuously evaluates national performance against those allocations.

The platform combines historical emissions data, population trajectories, policy commitments, and scenario-based stress testing to construct a comprehensive climate-equity accounting system. Through this approach, countries can be classified as carbon debtors or carbon creditors depending on whether cumulative emissions exceed or remain below their allocated share of the global budget. Beyond historical accounting, the framework incorporates policy-gap diagnostics, structural overshoot analysis, and emerging risk modules. These extensions enable the assessment of how legacy industrial structures, fossil-fuel dependence, and future electricity-demand pressures, including potential AI-driven growth scenarios⁵, may influence overshoot timing and future climate liabilities. The Carbon Budget Tracker is designed as a modular and extensible research platform. While the current implementation employs an Equal Per-Capita allocation methodology as a baseline equity framework, alternative allocation rules based on capability, historical responsibility, or hybrid burden-sharing approaches can be incorporated within the same computational architecture.

Research Question

To what extent have countries exceeded their fair share of the remaining global carbon budget, and what levels of carbon debt, policy misalignment, and future drawdown obligations emerge under alternative Paris-aligned pathways?

Additionally, how do structural development characteristics and emerging technological pressures influence overshoot timing and the remaining emissions runway available to individual countries?

2. Mathematical Framework & Core Metrics

The allocation core downscales the global remaining carbon budget into sovereign target boundaries using an equal per-capita equity principle.

2.1 System Boundaries, Allocation Rules, and the "Justice Score"

The downscaling of global atmospheric space to national boundaries requires making explicit normative choices regarding equity, baseline chronology, and capability. The *Carbon Budget Tracker* adopts an Equal Per-Capita (EPC) allocation rule applied to the remaining global carbon budget from a fixed baseline year, t_0 . While alternative burden-sharing frameworks exist, such as capability-based approaches (GDP-weighted resilience) or grandfathering methods (proportional to historical infrastructure footprint), the EPC framework treats the global atmospheric commons as a shared resource with equal extraction rights for all individuals.

To convert a nation's cumulative consumption pathway into a scannable normative metric, the platform constructs a composite Justice Score (JS_i), bounded between 0 and 100. This metric is a function of the Budget Used Percentage (BU_i), penalizing overexploitation while rewarding carbon storage or budget preservation:

$$JS_i = \begin{cases} 100 & \text{if } BU_i \leq 100 \\ \max(0, 100 - \alpha(BU_i - 100)) & \text{if } BU_i > 100 \end{cases}$$

Where α is a structural scaling sensitivity coefficient that dictates the rate of score decay as a nation accumulates carbon debt. For example, under current parameters, the United States has consumed 364% of its allocation, resulting in a severely decayed Justice Score of 16/100, while Germany sits at 204% consumption with a Justice Score of 61/100. Conversely, nations operating within their carbon budget baseline automatically retain a maximum score of 100/100, serving as global carbon creditors.

2.2 Dynamic Per-Capita Allocation

Let $B_{\text{global}}(t_0, \rho)$ represent the remaining global carbon budget designated by the IPCC AR6 at baseline year t_0 to limit global warming to a given threshold (e.g., 1.5°C) at a specified probability vector ρ (e.g., 50%). The equitable fair-share allocation A_i for a sovereign state i at time t is formulated as a function of its global population share:

$$A_i(t) = B_{\text{global}}(t_0, \rho) \times \left(\frac{P_i(t)}{P_{\text{global}}(t)} \right)$$

Where $P_i(t)$ represents national population and $P_{\text{global}}(t)$ represents global population.

2.3 Carbon Debt and Budget Used Percentage

The cumulative historical emissions of a country since the baseline year are balanced against this allocation to determine its current atmospheric liability. The Budget Used Percentage (BU_i) is computed as:

$$BU_i(t) = \left(\frac{\sum_{\tau=t_0}^t E_i(\tau)}{A_i(t)} \right) \times 100$$

Where $E_i(\tau)$ represents production-based annual CO₂ emissions in megatons (Mt CO₂). The absolute Carbon Debt (D_i) is expressed as:

$$D_i(t) = \sum_{\tau=t_0}^t E_i(\tau) - A_i(t)$$

- $D_i(t) > 0$: Indicates a **Carbon Debt** (systemic atmospheric overshoot).
- $D_i(t) < 0$: Indicates a **Carbon Surplus** (the country operates as a net climate creditor).

2.4 Policy Gap and Temporal Runway

To evaluate policy alignment, the engine determines the temporal runway (Y_i) remaining under current emissions trajectories, alongside the Policy Gap (G_i), which represents the disconnect between mathematical realities and net-zero targets:

$$Y_i(t) = \frac{A_i(t) - \sum_{\tau=t_0}^t E_i(\tau)}{E_i(t)}$$

$$G_i = t_{\text{pledge}} - t_{\text{overshoot}}$$

Where t_{pledge} is the self-declared NDC net-zero year, and $t_{\text{overshoot}}$ is the exact historical or projected year where cumulative emissions exceed the fair-share allocation.

3. Data Engineering Pipeline and Architecture

To handle the extensive data required for historical analysis and real-time parameter changes, the platform uses an optimized Python pipeline.

[Data Ingestion: OWID, WB, IPCC] —► [Index Harmonization via ISO-3] —► [Vectorized Equity Engine] —► [Reactive Streamlit Frontend]

3.1 Data Sources & Multi-Source Joining

The data processing engine programmatically integrates and cleans four primary data streams:

- **Emissions Records:** The framework utilizes the comprehensive *Our World in Data (OWID)* CO2 and greenhouse gas dataset, which aggregates the historical fossil fuel and industrial emissions registries curated by the Global Carbon Project^{6 7}. Spanning a temporal framework from 1750 to the present, the ingestion pipeline automatically resolves spatial anomalies, harmonizes split historical territories, and applies validation masks filtered by standard ISO-3 country codes.
- **Demographics:** Historical population data, demographic growth curves, and real GDP matrices are systematically pulled from the *World Bank World Development Indicators (WDI)* database⁸. These inputs serve as the critical denominator for tracking per-capita allocation drift over century-long horizons.
- **Physical Budgets:** The absolute planetary ceiling constraints are built upon the physical science bases established by the Intergovernmental Panel on Climate Change (IPCC) Sixth Assessment Report⁹. To adjust for ongoing real-world degradation of atmospheric stock between formal assessment cycles, the engine incorporates dynamic remaining carbon budget trackers based on the peer-reviewed *Indicators of Global Climate Change (IGCC)* frameworks¹⁰. This allows the application to downscale real-time tracking from specific milestone baselines (e.g., evaluating remaining global capacities

against metrics like a 362 Gt remaining threshold for 1.5°C targets at a 50% probability from a specified temporal baseline onward).

- *Policy Constraints:* Self-declared national climate target timelines, Nationally Determined Contributions (NDCs), and net-zero commitment horizons are harvested and encoded directly from the *United Nations Framework Convention on Climate Change (UNFCCC)* NDC Registry¹¹. This database acts as the baseline policy timeline against which the engine measures future overshoot acceleration or containment.

3.2 Computational Optimization

To ensure real-time rendering on the interactive Streamlit frontend, the Python backend eliminates iterative row loops. It relies on vectorized *pandas* and *numpy* matrix calculations across multi-indexed keys ['iso_code', 'year']. This optimization ensures that global metrics, ranking tables, and scenario changes update instantly upon user request.

4. Empirical Findings & Global Carbon Debt Diagnostics

4.1 The Asymmetry of Global Responsibility

When evaluated through a per-capita equity lens, the global carbon account exhibits a profound structural imbalance (**Fig. 1**). The engine determines that 51 countries have completely exhausted their fair-share atmospheric allocations, while only 27 countries remain net climate creditors. The top three historical debtor states, the United States (+136.1 Gt), the Russian Federation (+32.3 Gt), and Japan (+19.3 Gt), account for approximately 60% of all positive global carbon debt. Conversely, India functions as the world's primary climate creditor, maintaining 151.4 Gt of unused, equitable fair-share atmospheric capacity, having utilized only 26% of its fair share budget. To bridge macro-data layers with interactive policy evaluation, the platform centralizes these metrics into an open-access sorting matrix, compiled directly from vectorized country indices.

Global Carbon Debt Leaderboard

Countries with the highest historical fair-share overshoot.

View All >

ISO	COUNTRY	DEBT (GT)	BUDGET USED
USA	United States	136.1	363%
RUS	Russian Federation	32.3	228%
JPN	Japan	19.3	188%
DEU	Germany	14.7	204%
CAN	Canada	12.7	321%
SAU	Saudi Arabia	11.5	397%
KOR	Korea, Rep.	8.9	206%
AUS	Australia	8.8	337%
GBR	United Kingdom	6.3	159%
CHN	China	6.0	102%

Figure 1: Global Overview Tab - Core Leaderboard Engine. A clean capture of the web dashboard interface component. This module showcases the active "Global Carbon Debt Leaderboard" running under a 1.5°C threshold at 50% probability. It maps absolute historical overshoots alongside dynamic "Budget Used" constraints, highlighting the stark mathematical divide between debtor nations and global climate creditors.

4.2 Country Profile: Germany (DEU)

Isolating Germany provides an informative look at advanced, industrial European economies. Germany's fair-share IPCC allocation is calculated at 14.1Gt, while its historical cumulative emissions stand at 28.8 Gt (Fig.2). This creates a net carbon debt of +14.73 Gt (Budget Used = 204%). The engine identifies Germany's overshoot baseline year as 1990. Consequently, despite a steady downward curve in annual production emissions (E_{DEU}) and a substantial expansion in renewable energy market share over the past decades, Germany continues to accumulate absolute carbon debt. This historical lock-in results in a Justice Score of 61/100.

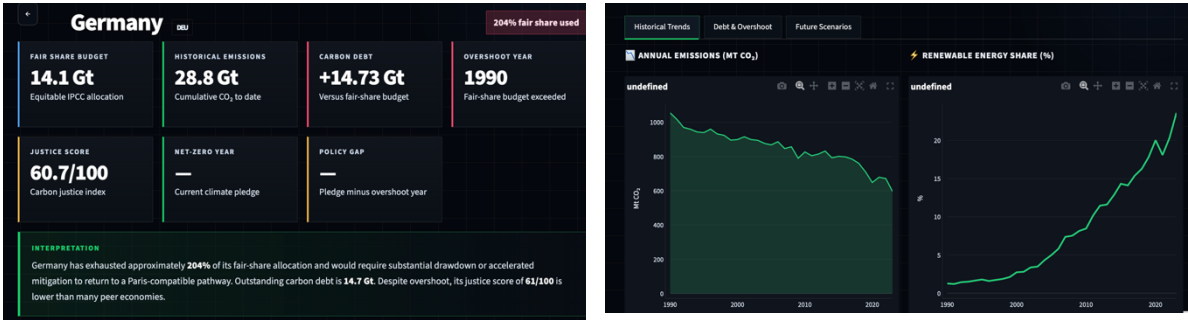


Figure 2: Sovereign Diagnostic Module and Trend Tracking (Germany Case Study). A composite screen capture from the user interface showcasing the deep-dive country profile view. The left panel isolates Germany's core resource metrics, including its absolute carbon debt (+14.73 Gt), structural overshoot baseline (1990), and its calculated Justice Score (60.7/100). The right panel provides empirical trend verification, illustrating the decoupling between a structural decline in annual production emissions ($Mt\ CO_2$) and an exponential increase in renewable energy market share (%) from 1990 through the present.

5. Policy Gap & Risk Quadrant Analysis

The tracker categorizes national performance by cross-referencing absolute carbon debt against temporal policy gaps (**Fig. 3**). The average global policy gap stands at 28 years, with a maximum gap of 30 years across the most severe overshoot nations.

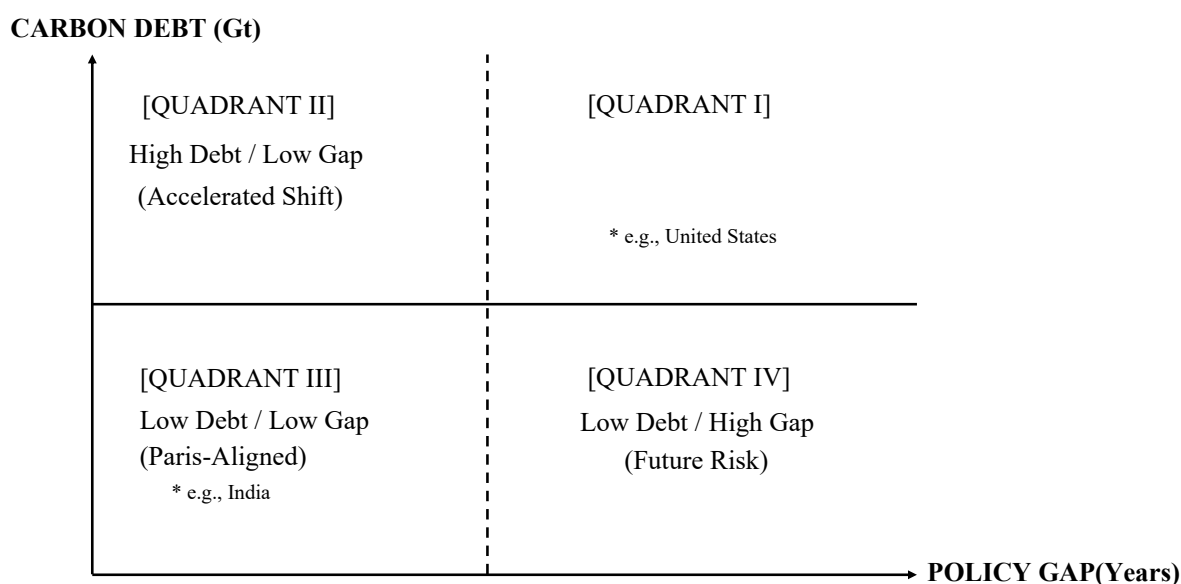


Figure 3: Global Climate Responsibility and Policy Alignment Matrix: This matrix categorizes national mitigation pathways by mapping absolute cumulative Carbon Debt (expressed in Gigatons, Gt) on the vertical axis against the calculated Policy Gap (expressed in years) on the horizontal axis.

- *Quadrant I: High Debt / High Gap (Highest Concern):* States displaying high structural carbon debt coupled with long delays before scheduled carbon neutrality. The United States anchors this quadrant with a severe policy gap and an exceptionally low Justice Score (16/100).
- *Quadrant II: High Debt / Low Gap (Accelerated Shift):* Legacy industrial economies that overshoot their budgets early but have implemented short carbon-neutrality target windows to limit further debt accumulation.
- *Quadrant III: Low Debt / Low Gap (Paris-Aligned):* Creditor nations whose cumulative pathways fall safely within their equity parameters. India represents the benchmark for this quadrant, with a Justice Score of 100/100 and a "Within Fair Share" designation.
- *Quadrant IV: Low Debt / High Gap (Future Risk):* Developing economies whose historical debt is negligible but whose long-term net-zero pledges lack the structural binding to prevent eventual budget overshoot under business-as-usual industrialization.

6. Advanced Stress-Testing: Overshoot Drivers and AI Scenarios

6.1 Structural Decomposition of Overshoot

The platform features an advanced analytics component that identifies why specific nations exceeded their budgets. As illustrated in the backend diagnostic outputs, Germany’s overshoot is strongly linked to historical industrial emissions rather than unusually high current fossil dependence. Germany's Industrialization Age is calculated at 231 years (70.5 years above the global median). This systemic lock-in acts as a powerful driver of overshoot. Meanwhile, its modern fossil share stands at 76.0%, which is actually 10.3% below the global median. This confirms that budget exhaustion is primarily an accumulation of historical structural inertia rather than contemporary policy failure.

6.2 Digital Transition Vulnerability: AI Sensitivity Engine

A core innovation of the tracker is its capability to model emergent systemic risks, specifically evaluating the impact of rapid digitization and data center infrastructure expansion on finite sovereign carbon runways. The platform stress-tests national climate accounts against three progressive digital infrastructure growth tiers: +10%, 25%, and +50%, grid electricity demand shocks. When isolating the primary case study country, Germany, the diagnostic engine uncovers a significant temporal vulnerability to tech-driven grid expansions. Under the worst-case 50% infrastructure surge scenario, Germany's baseline carbon debt is projected to expand significantly from its current 14.7 Gt up to 23.9 Gt (Fig. 4). This represents a sharp +63% explosion in cumulative carbon debt growth, which translates to an absolute liability expansion of +9.2 Gt over the baseline trajectory.

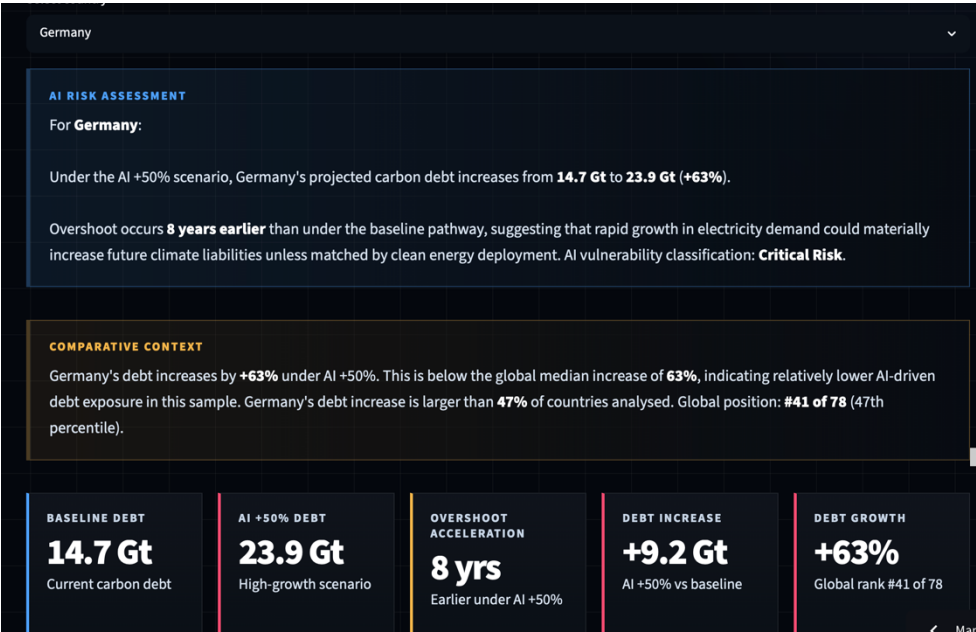


Figure 4: AI-Driven Digital Stress-Testing Diagnostic Panel (Germany View). Screen capture of the interactive predictive simulation output. The platform computes an absolute carbon debt increase from 14.7 Gt to 23.9 Gt (+63%) under peak digital energy stress conditions, triggering a forward-looking "Critical Risk" climate liability classification.

Crucially, this demand surge creates a severe temporal acceleration effect on the remaining policy runway. While Germany's historical 1990 baseline allocation breach remains fixed, the

predictive scenario tracker demonstrates that unmitigated data center power demands pull the country's forward-looking project boundary exhaustion timeline 8 years earlier than the baseline pathway. As illustrated by the predictive engine's trend tracking (Fig. 5), this causes the anticipated future threshold year to collapse precipitously from a baseline policy milestone of 2034 down to 2026 under peak +50% digital energy conditions.

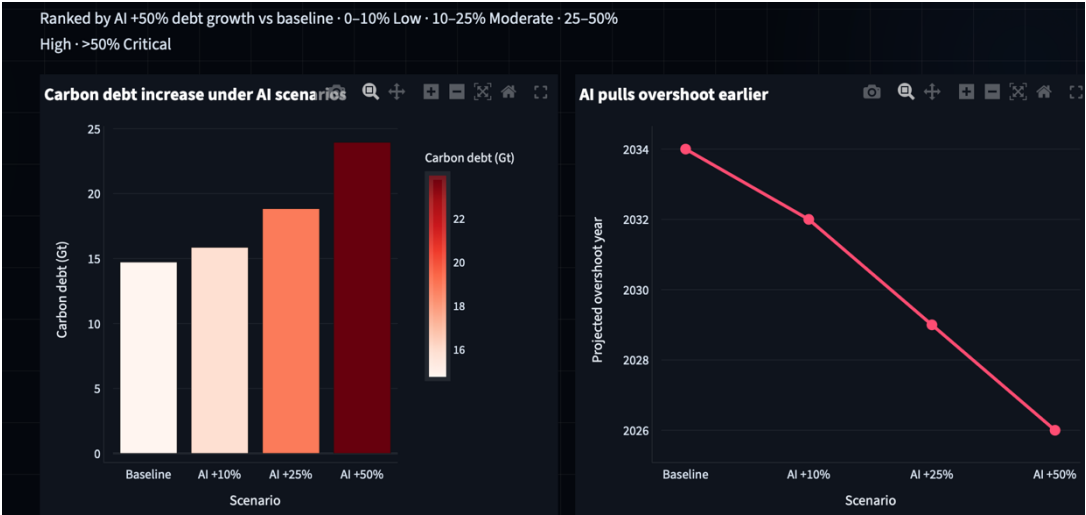


Figure 5: Empirical Carbon Debt Accumulation and Timeline Drift Plots. Visual capture of the backend plotting modules. The categorical bar chart (left) highlights the incremental scale-up of carbon debt across the +10%, +25%, and +50% computing tiers. The corresponding timeline line plot (right) maps the mathematical contraction of Germany's target runway, with the projected overshoot timeline dropping from 2034 to 2026.

To put these findings into a macro perspective, the tracker contextualizes Germany's performance against the broader global dataset (Fig. 6). Germany's debt growth rate of +63% sits exactly 1 percentage point below the global median growth rate of +64%, yielding a relative variance of -1 percentage point. This places Germany at an AI vulnerability rank of 41 out of 78 countries globally, anchoring it in the 47th percentile for relative digital transition risk exposure.

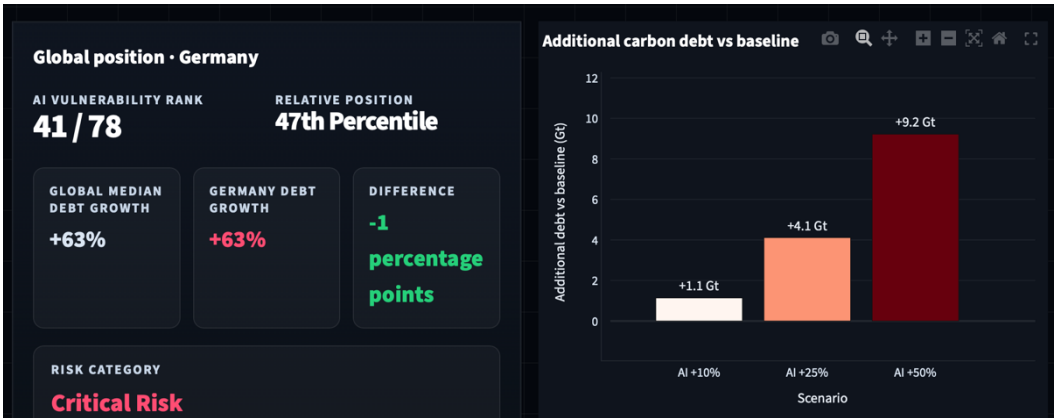


Figure 6: Global Position Comparison Module. A snapshot of the comparative analytics UI components (Screenshot 2026-06-17 at 10.35.15.png). The layout benchmarks Germany's absolute expansion against global boundaries, noting that while its growth rate closely tracks the global baseline median, the sheer scale

of the expansion (+9.2 Gt of additional carbon debt) positions the state as a major focus for digital-green policy coordination.

Even though Germany sits near the global median in terms of percentage growth, the physical volume of its liability expansion (+9.2 Gt) confirms that advanced industrial economies possess narrow tolerance margins for sudden energy infrastructure shocks. If computing loads expand without an immediate, equivalent deployment of dedicated post-fossil baseload capacity, the digital transition will directly cannibalize the remaining domestic carbon runway. This empirical finding shows that future iterations of international climate governance can no longer treat industrial digitization and carbon accounting as disconnected regulatory domains.

7. Limitations and Methodological Nuance

The empirical findings presented by the platform, including the calculation that 51 nations have completely exhausted their fair-share allocations, depend on specific choices within the model's design. To maintain academic credibility, these results must be interpreted as one specific equity lens rather than a definitive, singular measure of sovereign liability.

7.1 Baseline Year Sensitivity

The temporal anchor for this framework is set at a modern baseline using the remaining global carbon budget specified in the IPCC AR6. Shifting the baseline year changes the distribution of responsibility:

- *Historical Back-Casting (e.g., 1750 or 1990)*: Incorporating cumulative historical emissions from the start of the Industrial Revolution significantly increases the calculated carbon debt of legacy industrial nations (such as Germany and the United States).
- *Modern Forward-Looking Base (Current Framework)*: By focusing on the *remaining* budget (e.g., 362 Gt for at 50% confidence from 2025 onwards), the tracker offers a conservative estimate of legacy debt, evaluating how countries manage their shared runway from this point forward.

7.2 Alternative Equity Principles and Burden-Sharing Frameworks

The choice of an Equal Per-Capita (EPC) allocation rule represents a strict egalitarian approach to the global commons. In international environmental law and the principle of Common But Differentiated Responsibilities (CBDR), alternative allocation frameworks are frequently proposed:

- *Capability-Based Frameworks*: Allocating larger budget shares to nations with lower Gross Domestic Product (GDP) per capita or lower Human Development Index (HDI) scores to protect developmental space.
- *Historical Responsibility (Grandfathering)*: Allocating budget shares based on historical emissions pathways, which typically favors industrialized economies by softening the rate of required structural change, though it conflicts with standard climate justice literature.
- *Historical Lock-In and Technology Diffusion*: As shown in the dashboard's *Drivers* module, Germany's overshoot is strongly tied to an Industrialization Age of 231 years. The model currently treats this long industrial history as a primary structural contributor to overshoot. However, it does not credit legacy economies for the historical development and international diffusion of low-carbon technologies, which could be integrated into future iterations.

7.3 Data Constraints and Emerging Digital Pressures

The analytical outputs are bound by the accuracy of public production-based carbon accounting datasets. The current configuration does not fully account for consumption-based trade flows (embodied carbon in imported goods), which can shift carbon liabilities from consuming developed nations to producing developing nations.

Furthermore, as illustrated by the frontend's AI Sensitivity Engine, modeling future runway acceleration under an AI-driven +50% electricity demand scenario relies on current national grid emission factors. If a country couples digital expansion with rapid clean energy deployment, the projected acceleration of its overshoot year may be mitigated.

8. Strategic Implications for Global Climate Governance

The empirical outputs of the *Carbon Budget Tracker* challenge the foundation of current international climate negotiations. By framing progress around absolute emissions reductions rather than cumulative carbon debt, current frameworks inadvertently validate atmospheric appropriation by historical overshooters.

For institutions dedicated to transformation and sustainability, like RIFS Potsdam, these findings emphasize several core priorities:

1. *Redefining NDC Ambition:* National pledges must be evaluated against a country's absolute budget exhaustion year rather than arbitrary target years like 2050. A 30-year policy gap should be classified as a critical structural failure rather than a compliant pathway.
2. *Quantifying Loss and Damage Accounts:* The carbon debt metrics computed here (+136.1 Gt for the US, +14.7 Gt for Germany) provide a transparent data foundation for operationalizing international loss and damage funds, linking financial liabilities directly to atmospheric resource depletion.
3. *Integrating the Digital-Green Transition:* As confirmed by the tracker's AI Sensitivity module, climate governance can no longer treat digitization and decarbonization as separate domains. Policy frameworks must actively account for the energy demands of digital infrastructure to preserve remaining regional runways.

9. Conclusion

This paper introduced the *Carbon Budget Tracker* as an open-science computational framework designed to translate global carbon budgets into country-level metrics of responsibility, overshoot, and policy alignment. By applying an equal per-capita allocation principle to the remaining global atmospheric capacity, the platform exposes profound asymmetries in the consumption of a finite global commons. The empirical analysis indicates that 51 countries have exhausted their fair-share allocations and now operate with positive carbon debt balances, led by the United States (+136.1 Gt), while major developing economies such as India remain significant atmospheric creditors (-151.4 Gt). These findings highlight the unequal distribution of carbon budget consumption and provide a transparent framework for assessing climate responsibility beyond conventional emissions accounting. Beyond historical accounting, the framework connects climate equity to contemporary governance challenges through the integration of policy-gap diagnostics, structural overshoot drivers, and forward-looking stress-testing modules. The results reveal a substantial disconnect between physical carbon-budget constraints and long-term policy commitments, reflected in an average policy gap of 28 years among overshooting nations. The analysis further demonstrates that overshoot in legacy industrial economies is strongly associated with historical infrastructure lock-in and long-term industrial development pathways. The incorporation of an AI Sensitivity Engine extends the framework into emerging sustainability challenges by examining how rapid growth

in digital electricity demand may influence future carbon liabilities and budget exhaustion trajectories.

The findings should be interpreted within the methodological assumptions of the framework. The results are derived from an Equal Per-Capita allocation approach applied to a modern remaining-carbon-budget baseline and therefore represent one specific equity lens rather than a definitive measure of national responsibility. Alternative burden-sharing approaches based on capability, historical responsibility, developmental need, or hybrid allocation principles would generate different distributions of carbon debt and climate obligation. Nevertheless, the framework demonstrates how these normative choices can be made explicit, quantified, and explored transparently. The modular architecture of the *Carbon Budget Tracker* has been intentionally designed for future expansion. Planned developments include the integration of consumption-based emissions accounting, embodied carbon trade flows, capability-weighted allocation methodologies, and additional digital-transition scenarios. These extensions will enable a more comprehensive assessment of climate responsibility across increasingly interconnected economic and technological systems. Ultimately, the *Carbon Budget Tracker* serves as a transparent and reproducible platform for sustainability research, climate policy evaluation, and climate-equity analysis. By making carbon-budget calculations publicly accessible through an interactive dashboard and open analytical framework, the project supports evidence-based discussion on responsibility, fairness, and policy alignment under finite planetary boundaries. The framework offers a practical tool for exploring how physical limits, historical accountability, and future transition pathways intersect within climate governance.

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